

Service Bulletin DMU14-2024-03

Recurrence of Fanuc alarm 300 (APC need ZRN) after spindle replacement on DMG Mori DMU 50.
Applies to DMU-14, DMU-22.

```
--- METADATA ---
document_id: PDM-SB-DMU14-2024-03
title: Service Bulletin DMU14-2024-03 – Alarm 300 after Spindle Service
domain: machine_docs
source_system: PDM
revision: Rev A
effective_date: 2024-03-22
owner: DMG Mori Service / M. Tanaka (site liaison)
part_numbers: []
operation_numbers: []
machine_ids: [DMU-14, DMU-22]
controller_models: [Fanuc 0i-TF]
fault_codes: [300]
materials: []
standards: []
supplier_ids: []
jurisdictions: []
classification: internal
---
```

1. Symptom

After spindle motor or amplifier replacement on the DMU-14 / DMU-22 platform, Fanuc alarm **300** (APC need ZRN) reappears at every controller power-on for the affected axes, even after a single manual reference return.

2. Root cause

The absolute position coder (APC) loses its zero reference whenever the encoder cable is disconnected with controller power off, or whenever the servo amplifier is replaced. A single reference return clears the alarm *for the current session*, but the controller's stored APC reference is not re-written until a full APC parameter rewrite is performed.

3. Corrective procedure (DMU-14 / DMU-22)

Perform this procedure with the maintenance technician present. Door interlock override **must not** be used. If the procedure requires opening any guard, follow the machine LOTO procedure first.

1. Power on controller. Acknowledge alarm 300 on all affected axes.
2. Mode → JOG. For each axis (X, Y, Z, B, C), manually jog approximately 100 mm from the soft-limit, then perform reference return.
3. Enter parameter mode (key switch position 3, password required).
4. Set parameter **PRM 1815.5 (APCx)** = 1 for each affected axis. This instructs the controller to commit the current position as the new APC reference.
5. Power-cycle controller. Confirm alarm 300 does **not** recur.
6. If alarm recurs after this procedure: APC battery voltage may be below 3.0 V — replace battery (DMG Mori P/N MR-J3BAT) and repeat from step 1.

4. Cross-reference

For general alarm-clearance procedure: **PDM-MAN-FANUC-01TF-001** §7.2 and §7.3. Note that alarm 300 is a position-feedback fault, **not** a tool-wear fault. Tool-condition limits for the DMU-14 Inconel 718 process are documented separately in **PLM-PS-INC718-0042**.

Document control / sign-off

Role	Name	Signature	Date
Document Owner	DMG Mori Service / M. Tanaka (site liaison)	(on file)	2024-03-22
Approver	DMG Mori Service / M. Tanaka (site liaison)	(on file)	2024-03-22